ENVIRONMENTAL HEALTH

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Definition and basic concept of environmental health

- **By the end of the lecture the student will be able to:**
  - **Identify Environmental health, and its components.**
  - **Recognize sources of environmental risks**
  - **Global strategy for environmental health**
  - **Identify criteria for healthful Housing & Housing standards**
  - **Summarize housing and its relation to health**
  - **Understand the nature, health effects and sources of environmental risks for Air**
  - **Recognize effect of air pollution on health**
  - **Identify global effect of air pollution**
  - **Understand the nature, health effects and sources of environmental risks for water**
  - **Summarize process of purification**

Environment health definition

- Environment is the external factor/factors present around man and has got an influence on the health of the human being.

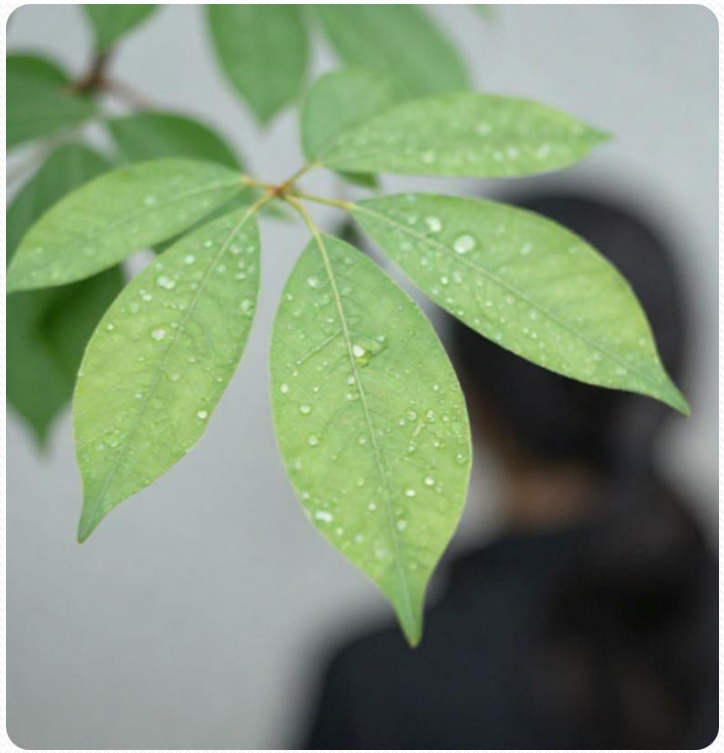
- According to ecologists, health is a state of dynamic equilibrium between man and his environment and when this equilibrium is disturbed, ill-health (disease) occurs.

- **Environment health definition: The term "environmental health" comprises the aspects of human health, including quality of life, that are determined by physical, chemical, biological, social, and psychosocial factors in the environment.**

- **Environmental health is the study and management of environmental conditions that affect the health and well-being of humans.**

Purpose of Environmental health

- The purpose of environmental health is to preventing human injury and illness and promoting well-being by
  - identifying and evaluating environmental sources and hazardous agents and
  - limiting exposures to hazardous physical, chemical, and biological agents in air, water, soil, food, and other environmental media or settings that may adversely affect human health



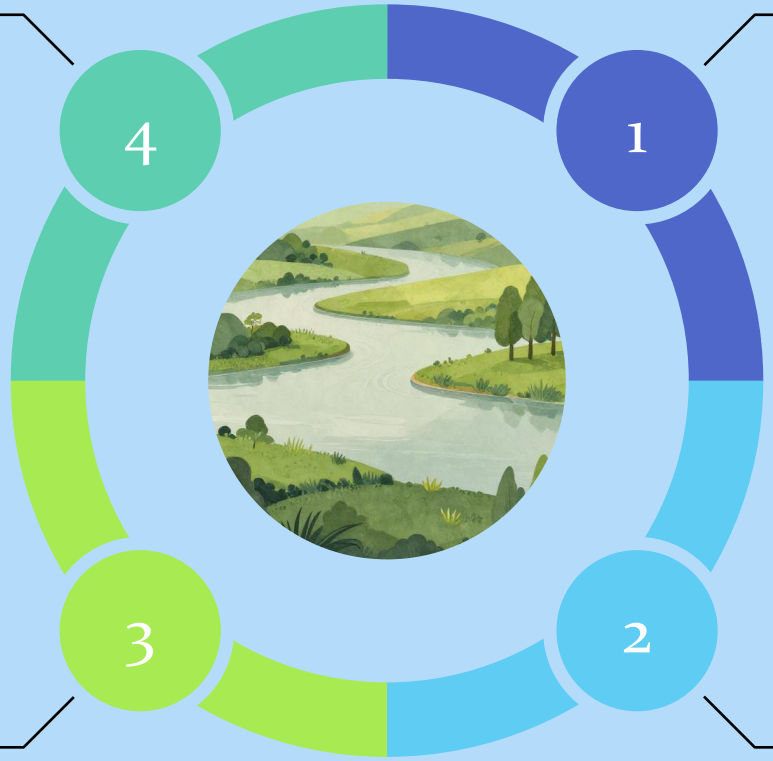
Environment components

CULTURAL ENVIRONMENT

This consists of knowledge, attitude, beliefs practices, traditions, culture, customs, habits, etc

PHYSICAL ENVIRONMENT:

This consists of non-living things and certain physical forces/ energy present around man.



This consists of occupation, literacy, income, religion, standard of living, lifestyle, availability of health services, etc.

SOCIAL ENVIRONMENT

This consists of living things around man. These are plants, animals, rodents, insects and microbes like bacteriae, viruses, rickettsiae, parasites, fungi, etc.

BIOLOGICAL ENVIRONMENT

Made with Twistly

Environmental sanitation Environmental sanitation

- The term environmental sanitation is defined by WHO as, ‘the control of all those factors in man’s physical environment, which exert a deleterious effect on physical development, health and survival’.

The objectives of environmental sanitation

- Improve quality of life.
- Prevent disease via clean homes, neighborhoods, communities.
- Promote hygiene (food, housing, workplaces).
- Control pests (insects, rodents).
- Encourage healthy behavior at individual, family, community levels.

Environmental risks (Hazards)

- **Physical**: heat, noise, vibration, radiation, pressure.
- **Biological**: pathogens, parasites, allergens (pollens), natural toxins (plants, fish).
- **Chemical**:
  - ***Irritants***: corrosives, caustics → tissue damage, infections, cancer risk.
  - ***Fibrotic agents***: silica (silicosis), coal dust (black lung), cotton dust (byssinosis), asbestos (carcinogenic).
  - ***Asphyxiants***: CO, HCN, H_2S → interfere with oxygen uptake.

- *Allergens*: e.g., formaldehyde → immune reactions.
- *Neurotoxins*: insecticides (DDT, malathion, parathion).
- *Mutagens*: DNA damage → heritable effects.
- *Carcinogens*: cause malignant tumors.
- *Teratogens*: cause developmental abnormalities in embryos.

- **Psychological** – stress → anxiety, cardiovascular disease, ulcers.
- **Diet-related** – nutrient deficiencies (e.g., iodine), malnutrition, poor eating habits → interact with infections.

Anthropogenic (Human-Caused) Damage

- Air pollution.
- Water pollution.
- Soil pollution & erosion.
- Deforestation.
- Desertification.

WHO standards for Environmental global strategy

- Improving the performance of environmental public health programs.
- Strengthen environmental public health surveillance.
- Foster an environmentally responsible and resilient health sector.
- Promoting environmentally healthy and resilient communities.

THEMATIC AREAS

- Air Quality and Health
- Climate Change and Health
- Chemical Safety
- Water and Sanitation

Housing and rural health

- **Criteria for Healthful Housing & Standards**
  - **Adequate living space – prevent overcrowding.**
  - **Safe indoor temperatures – avoid extremes of heat and cold.**
  - **Structural safety – minimize injury hazards (stairs, flooring, electrical safety).**
  - **Accessibility – housing suitable for people with disabilities or functional impairments.**
  - **Sanitation & water supply – safe drinking water, hygienic waste disposal, proper kitchen and bathroom facilities.**
  - **Ventilation & air quality – reduce indoor pollutants, ensure airflow.**
  - **Protection from vectors & chemicals – minimize exposure to pests, toxic substances, and unsafe building materials.**

Housing and Its Relation to Health

- **Physical health:** Poor housing contributes to respiratory diseases (from dampness, poor ventilation), injuries (unsafe structures), and infectious diseases (from inadequate sanitation and crowding).
- **Mental health:** Overcrowding, insecurity, and poor-quality housing increase stress, anxiety, and depression.
- **Social well-being:** Adequate housing supports dignity, privacy, and community resilience.
- **Public health impact:** Improved housing reduces disease burden, enhances quality of life, and supports equity in health outcomes.

Rural housing

- Poor housing quality (overcrowding, lack of sanitation, unsafe structures) increases infectious disease, injuries, and respiratory illness.
- In rural settings, housing often reflects broader inequities limited infrastructure, unsafe water, and exposure to environmental hazards.
- Improving housing standards directly strengthens rural health outcomes and reduces inequities.

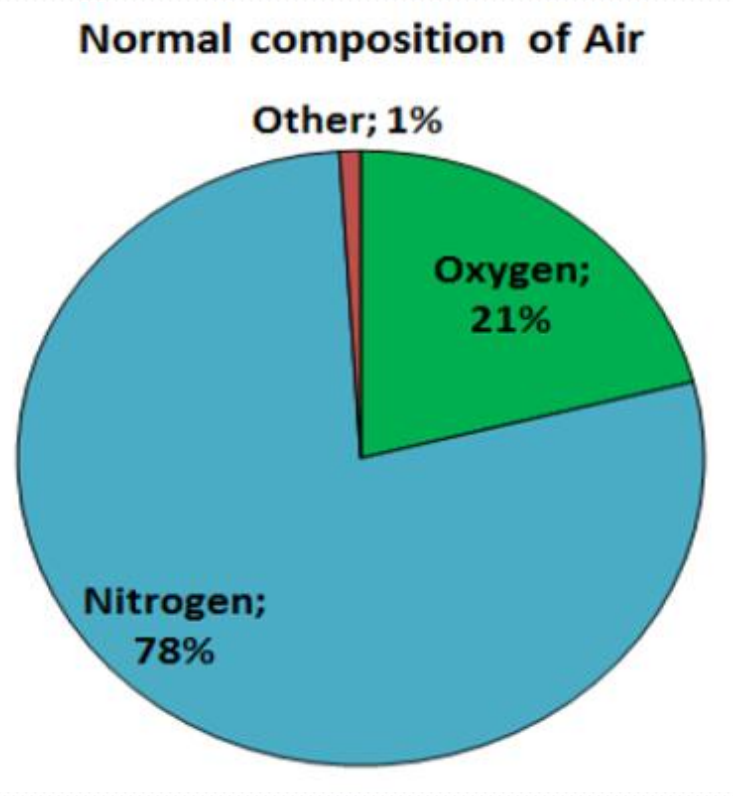
Standards of Rural Housing

- **Housing must provide adequate space to prevent overcrowding and disease spread.**
- **Houses should ensure structural safety, being stable and weather-resistant.**
- **Homes must maintain safe indoor temperatures and good ventilation for healthy air.**
- **Every household should have safe water supply and sanitation facilities.**
- **Housing must protect against vectors and harmful chemicals.**
- **Rural housing should be accessible and equitable, serving vulnerable groups and near essential services.**
- **Housing design should integrate with local climate, culture, and resources using sustainable materials.**

Air pollution

- **Air pollution is change in normal air constituents by introduction of chemicals, particulates, biological materials, or other harmful materials into the Earth's atmosphere, possibly causing disease, or damage to human being and other living organisms such as food crops, or the nature.**

- Under ordinary conditions, the composition of outdoor air is remarkably constant. This is brought about by certain self-cleansing conditions:
  - 1.Wind: wind dilutes and sweeps away impurities by its movement.
  - 2.Sunlight: it plays its role by oxidizing impurities and killing bacteria.
  - 3.Rain: It removes the suspended and gaseous impurities.
  - 4.Plant life: It generates oxygen.



Indicators of Air pollution:

- Sulfur dioxide
- Carbon monoxide
- Nitrogen dioxide
- Ozone
- Suspended particulates
- Carbon black.

  - The level of such pollutants are changeable according to the season as they are affected by air temperature and humidity.

Sources of air pollution:

- **Natural Sources** –Volcano, forest fire, dust storms, oceans, plants and trees
- Sources initiated **by human beings**:
  - **Industrial sources:** Factories of iron and steel, paper, cement, fertilizers, thermal power plant, petroleum refineries, etc.
  - **Incineration** or open firing.
  - **Domestic sources**: Burning of firewood, kerosene oil, coal, etc.
  - **Vehicular sources**: Motor vehicles, railways, ships, aeroplanes, etc.
  - **Miscellaneous**: Tobacco smoking, nuclear explosions, forest-fires, volcanoes, burning of refuse, dust-storm, ocean spray, etc.

EFFECTS OF VITIATED AIR

A. Health Effects

1. Immediate or short term effect:

- Irritation to the eyes, nose and throat,
- Upper respiratory infections such as bronchitis and pneumonia.
- Other symptoms can include headaches, fatigue, nausea, and allergic reactions.
- Aggravation of the medical conditions of individuals with asthma and emphysema.
- In the great "Smog Disaster" in London in 1952, four thousand people died in a few days due to the high concentrations of pollution.

2. Long term effect:

- Chronic bronchitis with persistent cough and expectoration.
- Asthma due to presence of allergens.
- Kidney diseases due to inhalation of nephrotoxic fumes.
- Heart diseases secondary to increase of pulmonary pressure.
- Cancers due to presence of initiator and promoter carcinogens.
- Abortions and other reproductive hazards as congenital anomalies.
- Depression and other psychological disorders.

Global effects due to air pollution

- **Acid Rain –** Caused by sulfur dioxide and nitrogen oxides; damages vegetation, structures, and can affect areas far from the pollution source.
- **Greenhouse Effect & Global Warming** – From fossil fuel combustion; leads to rising sea levels, melting ice, ecosystem imbalance, and biodiversity loss.
- **Ozone Layer Depletion** – Allows harmful ultraviolet radiation to reach Earth, increasing risks to skin, eyes, and overall health.
- **Radioactive Disasters** – Nuclear accidents release radiation; health risks and evacuation depend on radiation type and half-life.
- **Dioxin Pollution** – Toxic chemicals from incineration, smoking, exhaust, and smelting; linked to immune and reproductive damage, diabetes, cancer, and skin conditions (chloracne).

Prevention and control of air pollution)

- **Energy Transition**
  - Replace fossil fuels with clean energy (solar, wind, hydro, nuclear).
  - Promote energy efficiency in homes, industries, and transport.
- **Industrial Controls**
  - Enforce strict emission standards for factories and power plants.
  - Use cleaner production technologies and pollution filters.
- **Transport Measures**
  - Expand public transport systems.
  - Encourage electric vehicles, cycling, and walking.
  - Reduce reliance on private cars.
- **Urban & Rural Planning**
  - Establish industries away from dense populations.
  - Increase green spaces and urban forests to absorb pollutants.
- **Household Interventions**
  - Promote clean cooking fuels (LPG, electricity, biogas) instead of biomass.
  - Improve ventilation in homes to reduce indoor pollution.

Safe drinking water

- **Safe water should be**
  - **Clear**
  - **Pleasant to taste (i.e. free from odor and color i.e. odorless and colorless)**
  - **Free from pathogens**
  - **Free from harmful chemical substances**
  - **Usable for domestic purposes.**

Sources of Water:

- **I. Rain Water:**
  - **usually collects dust, bacteria and other impurities from the atmosphere and from any surface on which it falls. Rain water contains no dissolved salts, so, it may dissolve metals. it should be kept in concrete tanks and distributed through special type of pipes coated from inside with good asphalt paint.**
- **II. Surface Water:**
  - **This includes water from**
    - **fresh water lakes**
    - **Rivers, canals and streams, and**
    - **distilled seawater.**
- **III. Underground Water:**
  - **The underground water *may get out through***
  - **Springs: These are natural yields of water.**
  - **Wells are artificially made to get out the underground water.**

Water pollution

- **Definition:**
  - **Water pollution occurs when harmful substances (chemicals, waste, microorganisms) contaminate rivers, lakes, oceans, or groundwater, degrading water quality and making it unsafe for humans, animals, and ecosystems.**

- **Effects of water pollution:**
  - **Human health: Causes waterborne diseases (cholera, diarrhea, hepatitis).**
  - **Ecosystems: Kills aquatic life, disrupts food chains, reduces biodiversity.**
  - **Economic impact: Affects agriculture, fisheries, and safe drinking water supply.**
  - **Environmental damage: Leads to eutrophication, oxygen depletion, and habitat loss.**

Prevention & Control of water pollution

- **Industrial regulation**: Treat wastewater before discharge; enforce emission standards.
- **Agricultural practices**: Reduce pesticide/fertilizer runoff; promote organic farming.
- **Waste management**: Proper disposal of sewage, plastics, and hazardous waste.
- **Community action**: Raise awareness, conserve water, and avoid littering in water bodies.
- **Global cooperation**: Strengthen international agreements to protect shared water resources.

WATER PURIFICATION IN LARGE SCALE

- **Surface water is usually polluted and contaminated and is often unsafe and should not be used without purification.**
- **The process of purification aims at eliminating harmful, unpalatable or objectionable materials and removing the harmful bacteria.**
- **To supply large communities with potable sanitary water, large scale water purification plants are needed.**

Large scale Water Purification of Surface Water:

- The process of water purification in large-scale purification plants includes four main steps namely:
  - water intake,
  - sedimentation,
  - filtration and
  - disinfection.

Water intake:

- Intake pipes are placed upstream from the city and pollution sources, ensuring future expansion is considered.
- Protected zones: 500 m upstream and 150 m downstream must remain free of pollution.
- Pipes extend to the center of the river, midway between surface and bottom, avoiding floating debris.
- Diameter: 0.6–1 m, with a screen and one-way valve to prevent the water from getting out of the purification plant.

Sedimentation

- Water stored in large tanks for hours (4:8) to weeks (4 week), depending on demand.
- Coagulant (alum) added: reacts with bicarbonates to form aluminum hydroxide floc, trapping suspended matter.
- Alum concentration: 14–70 ppm, adjusted to impurity levels.
- Clear supernatant water exits from the top of the tank.
- Modern method: Dorr Clarifier – a rotating pillar with iron plates creates centrifugal movement, enhancing sedimentation efficiency.

Filtration:

- **Removes** remaining suspended matter and microorganisms.
- **Rapid sand filter** is most suitable for large-scale use:
  - Concrete tank (5–6 m diameter).
  - Base: branching metallic pipes with strainers, leading to exit pipe.
  - Layers: graded gravel (60–90 cm) topped with sand (60–90 cm).
  - Water head: about 100 cm above sand, controlled by inlet valve.
- **Surface film (biological layer)** forms at the top sand layer:
  - Contains suspended matter, traces of alum, algae, protozoa, and bacteria.
  - Develops in 15–30 minutes with alum traces.
  - Becomes denser over time, slowing filtration, so requires periodic washing or sand replacement.
- Rapid sand filters reduce bacterial content to about 5% of original load.

Disinfection:

- This is the last step in large-scale water purification.
- As indicated above, the water coming out of the rapid sand filter still contain bacteria and are not safe for human use unless it is disinfected.
- This is usually done by chlorine gas that is added to the water from big cylinders of liquid chlorine.
- A sufficient dose of 1 - 1.2 PPM. provides 0.1 - 0.2 PPM. of' residual chlorine after a contact time of 30 minutes.

Sanitary Standards of good drinking water:

1. Physical:

- Colorless, odorless, clear and tasteless.
- Drinking water should be free from turbidity and color.

2. Chemical:

- It should not be acid or strongly alkaline. It should be neutral or moderately alkaline at pH 8.00
- It should not contain an undue amount of solids. The maximum amount of solids should not exceed 500 parts per million.
- Hardness should not exceed 20 degrees of which not more than 4 degrees are permanent hardness.
- Hardness is defined as the soap-destroying power of water. It is classified into:
  - 1-Carbonate "temporary" is due to the presence of calcium and magnesium bicarbonates.
  - 2-Non-carbonate "permanent" is due to calcium and magnesium sulphates.

- **Chemical indicators of organic matter contamination should conform to the following standards:**
  - **It should be free from nitrogenous organic matter.**
  - **Compound organic matter should not more than 0.1 PPM.**
  - **Albuminoid ammonia, another first product of organic decomposition should be less than 0.01 PPM.**
  - **The Oxygen absorbed in 3 hours at 270 C (to oxidize the organic matter in the water) should be less than 3 – 4 PP.**
  - **Nitrites: should never be present since they are intermediate products of organic oxidation and their presence indicates recent and still going on pollution.**
  - **Nitrates are final products of organic oxidation. Their presence in water indicates past pollution. They should not exceed 1.0 PPM in surface water and 5.0 PPM in deep well water.**
  - **Nitrates may be present in deep well in higher concentration due to very ancient organic matter in the soil as in case of nitrate soil in Chilli.**

3. Biological

- **Bacteriological**:
  - **Total bacterial count:100 organisms /ml**
  - **Presumptive coliform: absent in 100 ml of filtered chlorinated water.**
  - **Ideally drinking water should not contain any microorganisms known to be pathogenic.**
  - **The primary bacteria is the coliform group, including both fecal and non – fecal organisms.**
  - **Typical example of the fecal group is E. coil and of the non – fecal group klebssiella aerogens.**

- **Bacteria that indicates fecal contamination of water;**
  - **Echerichia Coli:** normaly no coliform organism in 100 ml water.
  - Streptococcus faecalis: poorly survive in water and its presence indicates recent contamination with fecal matter.
  - **Colstridium welchi:** survives for long time and if present indicates remote contamination with fecal matter
- Drinking - water should be free from any viruses infectious to man.
- Drinking water should not contain any pathogenic intestinal protozoa. Rapid and slow sand filtration is effective in removing a high proportion of pathogenic intestinal protozoa.

4. Radiological -Drinking water should be free from any source of radiation.

THANK YOU